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CASE REPORT

A novel frameshift mutation in Allan-Herndon-Dudley syndrome

Zihao Liu[1] · Shuquan Zhao[1] · Jianyi Chen[1] · Longda Ma[1] · Qing Shi[1] · Yiwu Zhou[1]

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Abstract
Allan-Herndon-Dudley syndrome (AHDS) is a very rare, X-linked psychomotor disability syndrome with delayed myelination, almost exclusively affecting boys. We present a case of a 4-year-old boy with AHDS who was found cyanotic, with intermittent vomiting and paroxysmal convulsions about 4 h after his parents went out, and was then taken to the hospital, where he eventually died the next day. The autopsy revealed foreign bodies in the tiny bronchi and alveoli of the deceased, congestion, and punctate hemorrhage in multiple organs, consistent with the diagnosis of asphyxia. Compared with a normally developing 4-year-old boy, the deceased showed cerebral atrophy and cerebral edema, and Luxol Fast Blue (LFB) stain indicated delayed cerebellar, hippocampal, and basal ganglia development and myelination. A novel frameshift mutation c.584delG in the SLC16A2 gene was detected. Family lineage investigation showed that the mutation was also detected in the deceased's 8-year-old brother and biological mother. The present work enriches the profile mutations in SLC16A2 related to AHDS and emphasizes the importance of autopsy and postmortem genetic analysis in such cases.

Keywords Forensic pathology · Allan-Herndon-Dudley syndrome · SLC16A2 gene · Monocarboxylate transporter 8 · Myelination

Introduction

Solute carrier family 16 member 2 (*SLC16A2*) encodes the thyroid hormone (TH) transporter monocarboxylate transporter 8 (MCT8) located on Xq13.2. Mutations in SLC16A2 were first reported in a severe complex X-linked intellectual disability with abnormal thyroid hormone function test results [1]. *SLC16A2* was subsequently identified as the causative gene in Allan-Herndon-Dudley syndrome (AHDS) [2, 3]

Allan-Herndon-Dudley syndrome (AHDS, OMIM #300523) is a very rare, X-linked psychomotor disability syndrome with delayed myelination due to thyroid transport disturbances [2]. Patients show muscle wasting, spastic quadriplegia, dystonic movements, severe intellectual disability (intelligence quotient < 30), and lack of speech [4]. This condition was described in 1944 by Allan et al. and almost exclusively affected boys [2]. Abnormal thyroid hormone (TH) serum levels are a hallmark of the disease. Patients with AHDS usually have low free tetraiodothyronine (T4), high serum levels of free 3,5′,3′-triiodothyronine (T3), and normal or mildly elevated thyroid stimulating hormone (TSH) [1, 5]. Brain magnetic resonance imaging (MRI) commonly shows hypomyelination in the early years, with subsequent progression and no associated clinical improvement [6–8].

SLC16A2 is expressed in a variety of tissues and has a major role in the development of the central nervous system [9]. SLC16A2 mutations are estimated to be involved in up to 4% of X-linked intellectual disability syndromes [10]. Deficiency induces a lack of T3 entry in the brain, leading to

Zihao Liu and Shuquan Zhao contributed equally to this work.

✉ Yiwu Zhou
zhouyiwu@outlook.com

Zihao Liu
fyliuzihao@outlook.com

Shuquan Zhao
fyzsq123@126.com

Jianyi Chen
1454507129@qq.com

Longda Ma
longdamafm@outlook.com

Qing Shi
806496165@qq.com

[1] Department of Forensic Medicine, Tongji Medical College, Huazhong University of Science and Technology, No. 13 Hangkong Road, Wuhan 430030, People's Republic of China

central hypothyroidism, and in other tissues (muscle, heart, skin, bone, intestine), clinical signs indicate reduced thyroid hormone availability, accompanied by mild peripheral thyroid dysfunction [1, 11].

Here, we present a rare fatal case of Allan-Herndon-Dudley syndrome. A 4-year-old boy with AHDS slept at home with his head covered with a quilt and died 1 day after being rescued by his parents. We present the clinical and autopsy results of the case. In addition, a genetic testing was performed and a novel frameshift deletion in *SLC16A2* (exon3:c.584delG:p.Ser195Thrfs*72) was detected. The Sanger sequencing of the family showed that the deceased's brother and mother had the same mutation.

Case presentation

A 4-year-old boy was admitted to the hospital due to cyanosis, intermittent vomiting, and paroxysmal convulsion for approximately 4 h. Eight hours earlier, his parents left him at home to sleep, and then, 4 h later, his parents returned home and found his head covered with a quilt. Lifting the quilt, the patient's face was pale and his lips were cyanotic, and then, he was taken to the hospital for treatment. The patient was diagnosed with cerebral palsy at 2 months of age and presented with intellectual and motor developmental impairment, which was similar to the presentation of his 8-year-old brother, while their parents were normal. Until the age of 4, the patient is still unable to sit or crawl on his own, and his muscle strength is so weak that he can only lift his head and arms and legs slightly and cannot roll over. On admission, the patient's blood examination revealed multiple organ impairment and clotting disorder. On the next day, the patient suddenly lost consciousness and died. The hospital provided a diagnosis of sudden death, multiple organ failure, and severe malnutrition. To find out the cause of death, forensic autopsy was performed.

Autopsy findings and histopathological examinations

The deceased weighed 8.3 kg and was malnourished. External examination revealed no sign of trauma or injury except for several injection points.

No subscalp hemorrhage, temporal muscle hemorrhage, or skull fracture was observed. A 11.0-cm × 1.3-cm × 0.3-cm subdural hematoma was found on the surface of the left hemisphere of the brain (Fig. 1a). No evident brain damage was seen below the subdural hematoma (Fig. 1b). The brain weighed 850 g, the sulcus was shallowed, and the gyri was widened. Coronal section of the brain showed white matter dysplasia compared to a normal 4-year-old boy (Fig. 1c, d). Microscopically, the brain was edematous, the neurons were ischemic and hypoxic, the subarachnoid space of the right frontal lobe had spot hemorrhage, and diffuse intradural hemorrhage was found. The deceased had hypoplasia in multiple parts of the brain, such as smaller widths of the cerebellar lobes and the olivary nucleus of the medulla oblongata compared to control boys. Luxol Fast Blue (LFB) stain showed smaller myelin diameters in the brainstem, cerebellum, corpus callosum, hippocampus, and intracortical areas of the deceased compared to the control boy of the same age (Fig. 1e–l).

Punctate bleeding was on the surface of the heart. Histopathologically, cardiac conduction system examination revealed focal hemorrhage in the interstitial myocardium in the atrioventricular node region (Fig. 2). A small amount of foreign body was seen in the trachea (Fig. 3a, b). Microscopically, the lung showed congestion, foreign body with inflammatory cell infiltration in the pulmonary fine bronchi and alveoli (Fig. 3c, d, e), and edema, focal patchy hemorrhage in the alveoli (Fig. 3f).

Thyroid hormone examination in the postmortem serum indicated the following: free triiodothyronine was 7.24 ↑ pmol/L (3.1–6.8 pmol/L), free tetraiodothyronine was 17.00 pmol/L (12.00–22.00 pmol/L), and thyroid stimulating hormone was 5.58 ↑ uIU/mL (0.27–4.20 uIU/mL). The weight, height, and BMI of the deceased and his 8-year-old brother showed below the standard values for children of the same age, indicating developmental delay. The deceased, his brother, and mother all showed high FT3, low FT4, and normal or slightly high TSH, suggesting a disorder of thyroid hormone secretion and the possibility a genetic disease in the family. More information of this family is shown in Table 1. Toxicological analysis ruled out common poisons and the poisons causing convulsions and bleeding.

Genetic testing

Considering the need of the diagnosis of the cause of death and the possible usefulness of the treatment for the 8-year-old brother who had similar symptoms and the parents of the deceased, we performed genetic testing after consulting the family of the deceased. With the consent of the deceased's parents, a whole-exome sequencing was performed using heart blood sample collected from the patient.

Genomic DNA was extracted from the heart blood by the kit instructions (MagPure Buffy Coat DNA Midi KF Kit, MAGEN). The gDNA was broken into fragments of 100 ~ 500 bp using fragmentation enzyme kit (Shearing Enzyme premix reagent, ENZYMATICS), and then magnetic bead (Vahtstm DNA Clean Beads, VAZYME) was used to collect 200 ~ 300 bp fragments. After repairing ending, "A" base was added at 3′ overhangs, and then, after

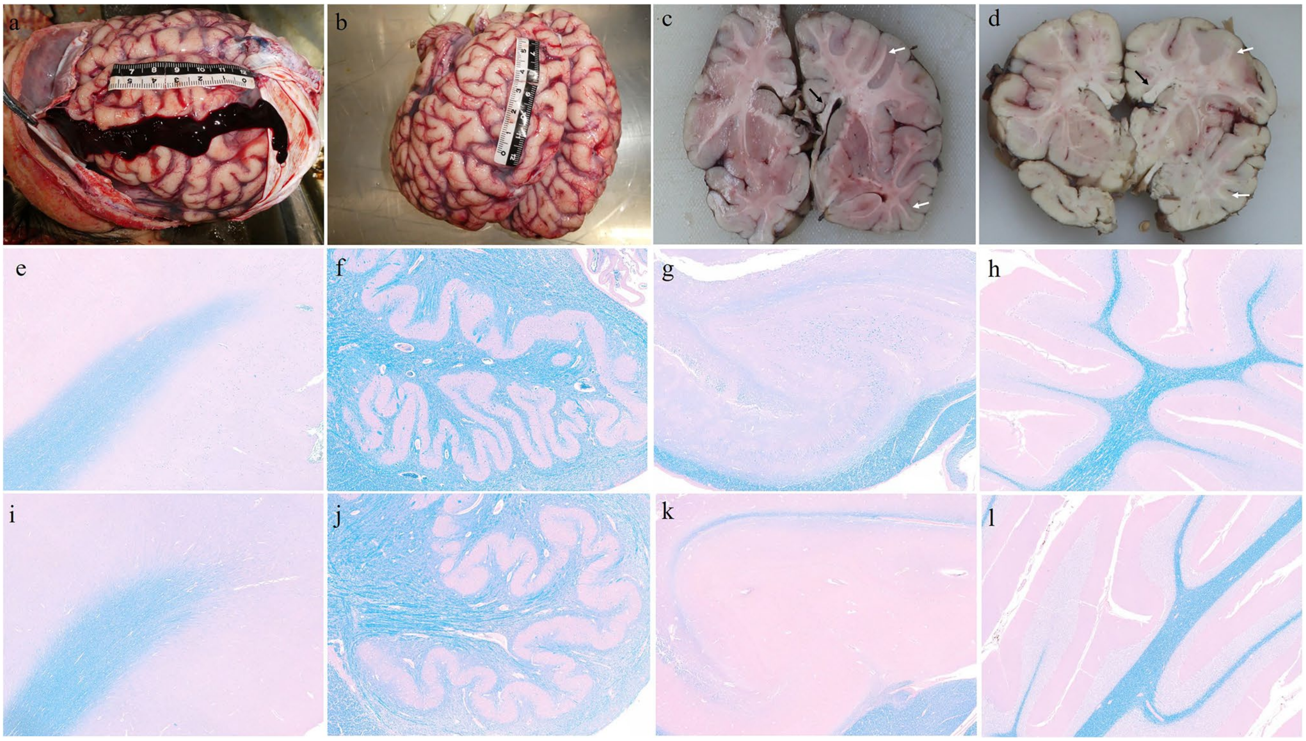

Fig. 1 Gross and microscopic of the brain. **a** A 11.0-cm × 1.3-cm × 0.3-cm subdural hematoma on the surface of the left hemisphere of the brain; **b** no brain damage below the subdural hematoma; **c** coronal section of the brain of the deceased in this case showed white matter dysplasia; **d** coronal section of the brain in control boys of the same age (**e**, **f**, **g**, **h** the deceased in this case; **i**, **j**, **k**, **l** control boys of the same age); **e**, **i** compared to control boys with superficial white matter fine branches, the deceased in this case showed poorer developmental levels (LFB, 20×); **f**, **j** the medulla myelin diameter of the control boy was larger than that of the deceased in this case (LFB, 20×); **g**, **k** myelination was poor in the hippocampal penetrating fibers of the deceased compared to control boys (LFB, 20×); **h**, **l** the diameter of myelin in the cerebellum of the deceased was smaller, and the staining was lighter compared to that of the control boy (LFB, 20×)

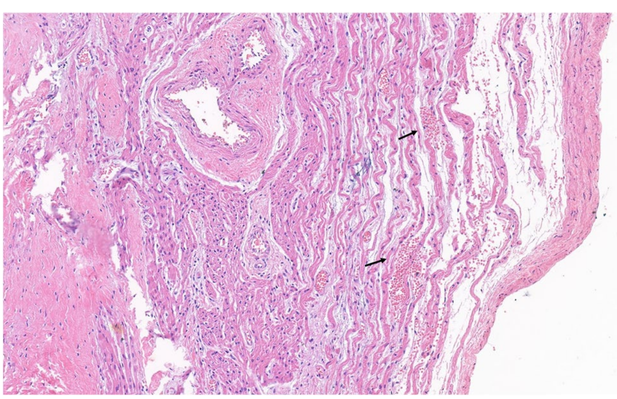

Fig. 2 Focal hemorrhage in the interstitial myocardium in the atrioventricular node region

LM-PCR and purification, a single individual DNA library was established. The library was enriched by array hybridization (KAPA HyperExome, ROCHE) at 65 °C for 16 h and then amplified after elution and capture. After detecting the degree of product enrichment by CaliperGX_1000Kit and BMG, the qualified products were pooled and quantified based on different library quantities, then the single-strand pooling library was prepared for circularization and made DNB, and finally sequenced on MGISEQ-2000. Data were interpreted in accordance with the relevant guidelines of American College of Medical Genetics and Genomics [12]. Variants were named according to the nomenclature recommended by the Human Genomic Variation Society.

Sanger sequencing was implemented using peripheral blood collected from the deceased's parents and brother; the mutation was detected in the family line by the Sanger sequencing platform.

In the present case, DNA sequencing revealed a novel frameshift deletion in *SLC16A2* (exon3:c. 584delG:p. Ser195Thrfs*72). This mutation did not exist in multiple population databases, including dbSNP (http://www.ncbi.nlm.nih.gov/snp), the 1000 Genomes Project (https://www.ncbi.nlm.nih.gov/variation/tools/1000genomes/), the International HapMap Project (http://http.ncbi.nlm.nih.gov/hapmap/), gnoMAD (http://gnomad.broadinstitute.org), and the NHLBI GO Exome Sequencing Project (http://evs.gs.washi

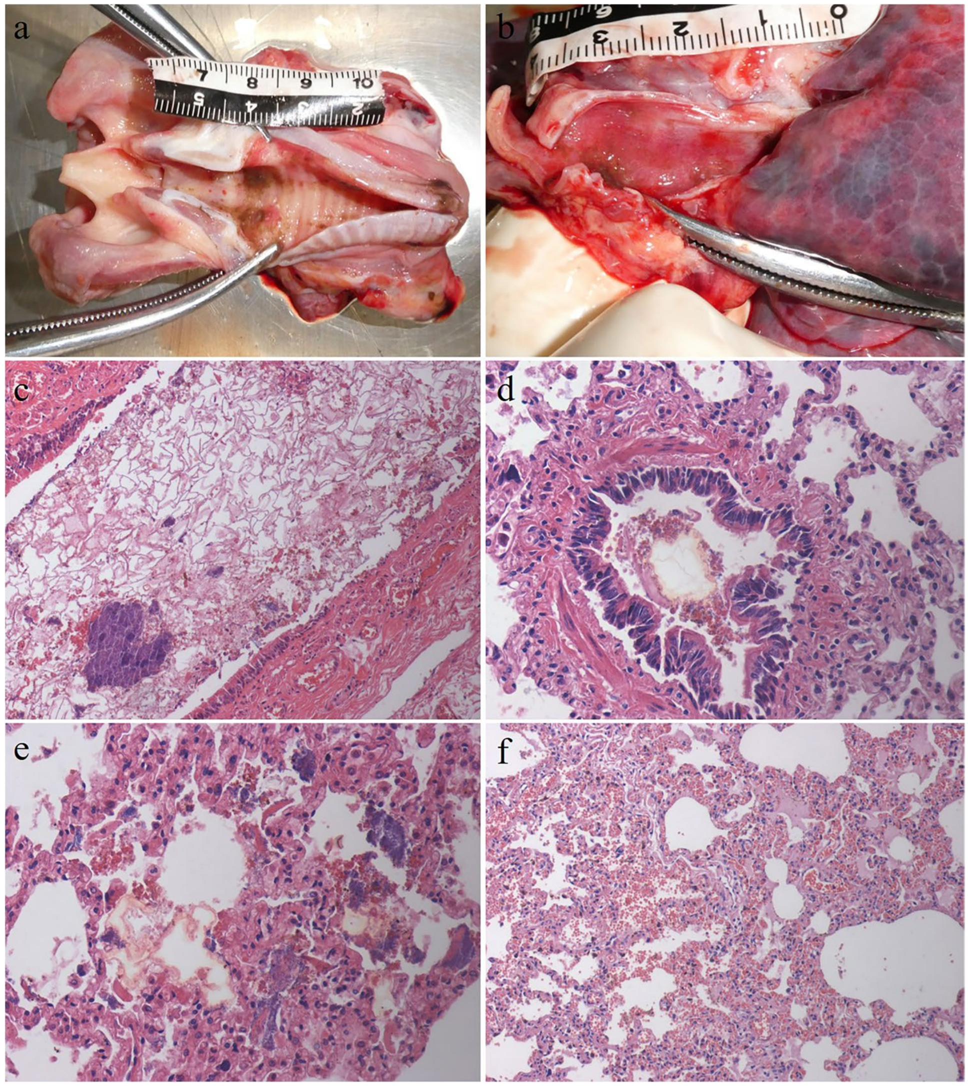

Fig. 3 Gross and microscopic of the lungs and bronchi. **a**, **b** A small amount of residual foreign body in the trachea and main bronchus; **c** vegetative foreign bodies and red blood cells in the bronchus (HE, 100×); **d** foreign bodies and red blood cells in the fine bronchus (HE, 200×); **e** foreign body in the alveoli with a small amount of neutrophil infiltration (HE, 200×); **f** pulmonary hemorrhage (HE, 100×)

Table 1 The other information of this family

	The deceased	Elder brother	Mother	Father
FT3 (pmol/L)	7.24↑	7.72↑	5.68	4.71
FT4 (pmol/L)	17.00	6.01↓	12.70	16.40
TSH (uIU/mL)	5.58↑	2.76	0.84	0.14↓
Age (y)	4	8	33	36
Weight (kg)	8.3 (<P_3)	12.5 (<P_3)		
Height (m)	0.91 (<P_3)	1.05 (<P_3)		
BMI (kg/m^2)	10.0 (<P_3)	11.3 (<P_3)		
Normal BMI (kg/m^2)	15.3	16.4		
Hypotonia	√	√		
Muscle hypoplasia	√	√		
Never walked	√	√		
Drool	√	√		
Inadequate head control	√	√		
Absent speech	√	√		
Esotropia	√	√		
Seizures	Onset at 6 months	×		

ngton.edu/EVS/). According to the suggestions of the American College of Medical Genetics (ACMG), the mutation is grouped as a "likely pathogenic variant," which is classified as PVS1 (null variant (frameshift) in genes in which loss of function (LOF) is a known disease mechanism) and PM2 (a from controls (or at extremely low frequency if recessive) in Exome Sequencing Project, 1000 Genomes or ExAC). The family's blood Sanger sequencing results showed that the deceased, his brother, and mother had the same mutation in *SLC16A2* (exon3:c. 584delG:p.Ser195Thrfs*72), while the father was normal (Fig. 4A, B).

Discussion

Allan-Herndon-Dudley syndrome (AHDS) or monocarboxylate transporter 8 (MCT8) deficiency is an X-linked psychomotor disorder syndrome caused by mutations in the SLC16A2 gene, which encodes the thyroid hormone (TH) transporter protein MCT8 [9]. MCT8 deficiency was linked to the altered TH levels, which was critical for the development of many organs, including the central nervous system (CNS) [13]. In the present case, the deceased had clinical signs of muscle wasting, hypotonia, severe intellectual disability, high 3,5′,3′-triiodothyronine (T3), low thyroxine (3,5,3′,5′-tetraiodothyronine, or T4) and slightly increased thyrotropin levels on thyroid hormones, and pathologically brain atrophy and delayed myelination, which were consistent with the diagnosis of AHDS. The same changes were seen in the thyroid hormone examinations of his brother and mother. The genetic testing confirmed the diagnosis of AHDS. Also, high levels of T3 in peripheral blood can lead to a hypermetabolic state and excessive depletion of skeletal muscle resulting in reduced muscle volume in AHDS patients, which may be the main reason for weight loss in these patients.

Respiratory, gastrointestinal, and orthopedic disorders were the most common complications in AHDS patients, which can lead to premature death [14]. A multicenter cohort study showed that 30% patients died in childhood, and the main causes of death in AHDS were pulmonary infection (18.8%), sudden death (18.8%), and aspiration pneumonia (9.4%) [15]. In the present case, foreign bodies in the fine and small bronchi and alveoli, visceral congestion, and punctate bleeding in multiple organs were seen, which was consistent with the diagnosis of asphyxia. On the other hand, a swollen brain and subdural bleeding tend to be automatically applied to the term "shaken baby syndrome." Nevertheless, there were no evident signs of abuse or unambiguous indicators of trauma in the deceased. In fact, the hypoxic dural vascular injury hypothesis has been proposed by Geddes that endothelial cells of meningeal vessels (mainly dural venous structures) are congested/dilated due to hypoxic injury, resulting in extravasation of blood and, therefore, blood accumulates in subdural spaces [16]. Upper airway obstruction due to vomit or food asphyxiation has been proposed as a causal mechanism [17]. In consequence, diffuse intradural hemorrhage in this case might be the source of subdural hemorrhage. Meanwhile, brain dysplasia due to AHDS caused by mutations in SLC16A2 may make venous structures in the dura mater more susceptible to hemorrhage. In addition, focal hemorrhage in the interstitial myocardium in the atrioventricular

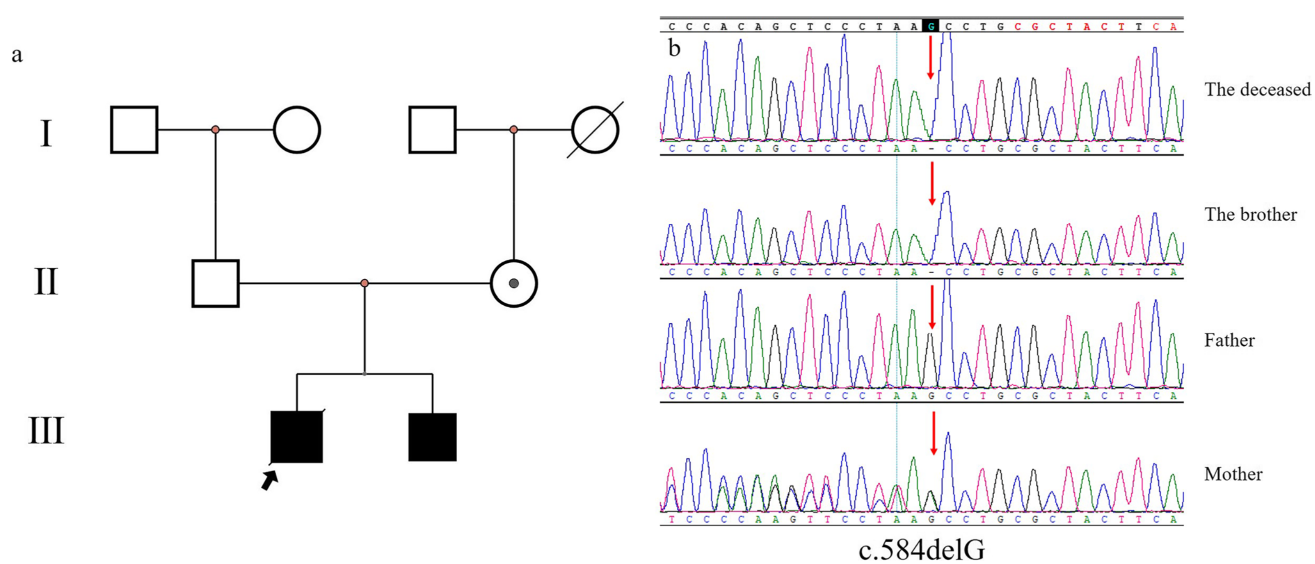

Fig. 4 The results of genetic testing. **a** Pedigree showing two boys with hemizygous mutation and a carrier mother; **b** chromatogram of SLC16A2 frameshift mutation

node region may also be caused by asphyxia and dysplasia or by extrathoracic compressions.

A critical developmental process modulated by TH is myelination [18]. In humans, the peak of myelination occurs mainly in the first 2 years of life and is largely complete by 5 years of age but continues in some regions (e.g., frontal cortex) until early adulthood [19]. Myelination follows a fixed temporal-spatial pattern, with axons in the brain stem forming myelin first, followed by those in the cerebellum, and finally the cerebral cortex [20, 21]. Impaired myelination is also a prevalent feature of patients with AHDS, most of whom have an abnormal white matter content in the first year after birth [14, 22]. In this case, the brain weight of the deceased showed brain atrophy compared to the standard pediatric table, which was also a sign of myelin dysplasia. Compared with a 4-year-old boy who died of hemolytic anemia with normal development, the deceased showed no significant abnormalities in brainstem myelin development, narrower molecular layers in the cerebellar lobes, poor myelination in the hippocampal penetrating fibers, and lighter myelin staining in the internal capsule and white matter of the basal ganglia. These changes are consistent with the fixed temporal-spatial pattern of myelin development but are below the developmental level of children of the same age. The observed alterations in hippocampal connectivity pathways could result in learning, memory, emotion, and motor control dysfunction. Cerebral atrophy and multiple sites of white matter hypoplasia may underlie features such as mental retardation, hypotonia of the trunk, and poor head control.

To our knowledge, the novel mutation c.584delG that was detected in our case has not been reported and is extremely minimal in the normal population. Sequence analysis showed a frameshift deletion of G at position 584, resulting in p.Ser195Thrfs*72. This novel variant was predicted to be pathogenic by SIFT, MutationTaster, and other functional prediction websites. The mutation was also detected in the deceased's biological mother without positive psychomotor disability. In fact, female carriers are commonly asymptomatic but can show mild symptoms of disease, like slight changes in thyroid hormone profile. The deceased's 8-year-old biological brother was also diagnosed with cerebral palsy a few months after birth and developed similar psychomotor disability. The brother started to show symptoms of convulsions at about half an age, usually occurring when he was asleep. In forensic practice, sudden death in children is common and often associated with genetic mutations. Therefore, postmortem genetic analysis is important to identify potential genetic disorders and links to causes of death [23, 24]. In addition, genetic testing of the family line can confirm the mode of inheritance of the disease and provide recommendations for treatment and re-birth for living relatives, highlighting the important value of forensic medicine as a medical discipline.

In conclusion, a novel frameshift mutation c.584delG in the SLC16A2 gene was likely pathogenic connected with AHDS. The deceased died of asphyxia due to inhalation of foreign bodies, and the pathology findings in the brain may contribute to explain the clinical symptoms. The present work enriches the profile mutations in SLC16A2 related to AHDS and emphasizes the importance of autopsy and postmortem genetic analysis in such cases.

Declarations

Informed consent Written informed consent for publishing this scientific report was obtained from the direct relative of the decedent in this case.

Conflict of interest The authors declare no competing interests.

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